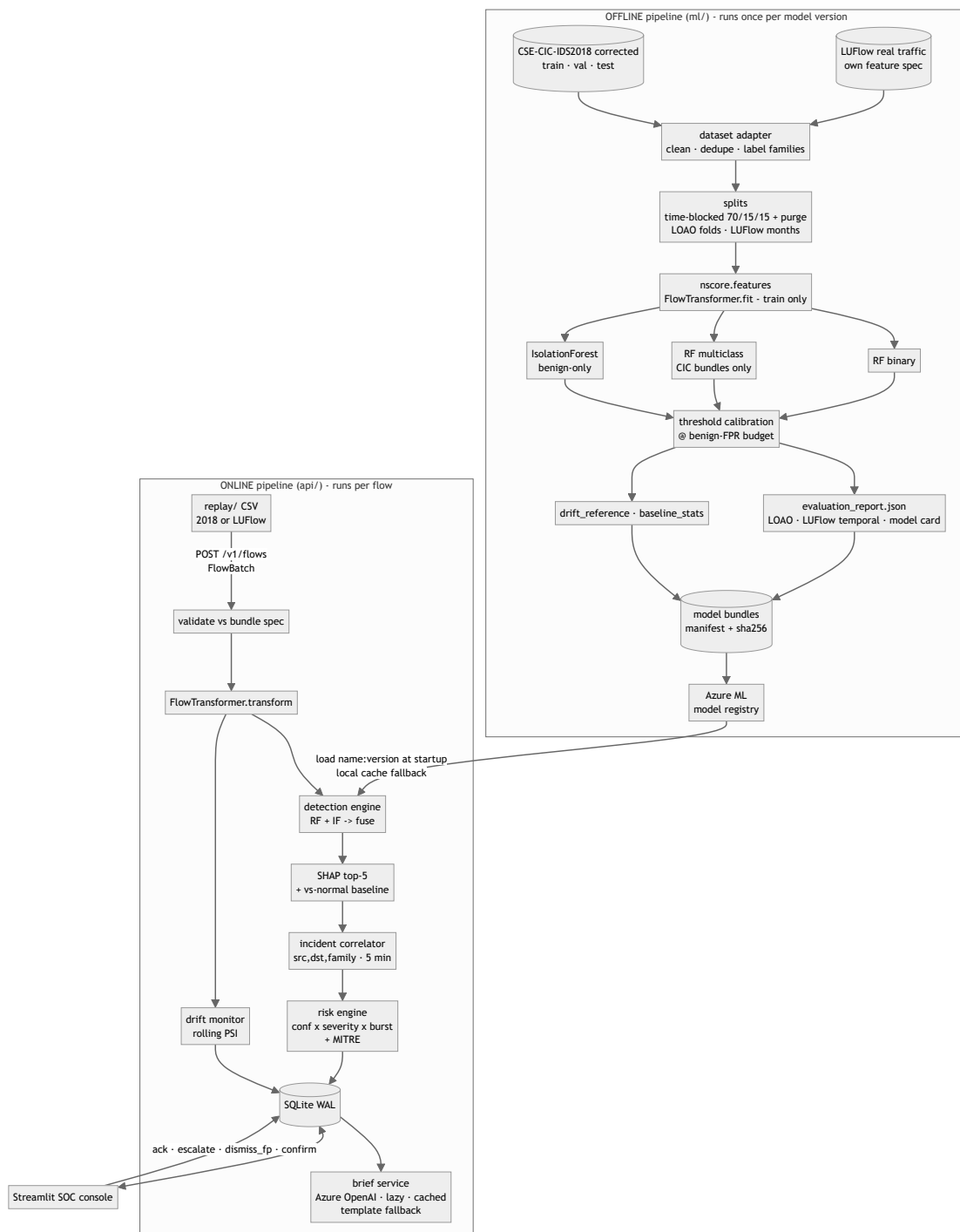


NetSentinel Final Architecture

One line: NetSentinel puts two detectors side by side. A Random Forest recognises known attack families. An Isolation Forest trained only on benign traffic flags anything that doesn't look normal. Their output is grouped into explained, **risk-scored incidents** for a SOC analyst. It's trained and tested on CSE-CIC-IDS2018 and shown working on real traffic (LUFlow). Nothing is ever auto-blocked.

1. System overview



Design principles

1. **Separate offline and online pipelines** (kept from Doc B). Serving never imports training-only dependencies.
2. **One shared core** (`nscore/`) for anything that must behave the same in training and serving: feature transform, fusion, risk engine, drift math, bundle loading. This prevents train/serve skew by construction.
3. **Spec-driven, so dataset-agnostic**. Every bundle carries its own `feature_spec.json`. The same code trains and serves a CIC-schema model (2018) and a LUFlow-schema model, with no dataset-specific branches downstream of the adapters.
4. **Contract first**. `nscore/contracts/` (schemas, policy, fixtures) is the only coupling between components, so five people can build in parallel.
5. **Honest by default**. Thresholds come from an FPR budget, every claim has a matching metric, and the limitations are written down.
6. **Demo-proof**. Every external dependency (Azure ML, Azure OpenAI, Wi-Fi) has a local fallback.

2. Repository layout

```
netsentinel/
├─ nscore/                shared core library (pip install -e .)
│   └─ contracts/         schemas.py · policy.py (risk engine) · feature_spec*.json · fixtures/ ← the contra
│   └─ features/          FlowTransformer (M1)
│   └─ detection/         fusion.py (M2)
│   └─ drift/             PSI + reference builder (M1)
│   └─ bundle/            manifest, packager, loader w/ Azure ML + cache (M2)
├─ ml/                   offline pipeline: data/ (adapters/) train/ evaluate/ explain/ registry/ (M1, M2)
├─ api/                  FastAPI service: app/ (routers, services, repo) tests/ (M3, M5 brief)
├─ dashboard/            Streamlit SOC console: app.py, pages/ (M4)
├─ replay/               replay engine + scenarios/*.yaml (M5)
├─ infra/                docker-compose, Azure provisioning scripts (M3)
├─ notebooks/           EDA + experiments (never imported by code)
├─ data/                 gitignored: raw/ processed/ splits/ replay/
├─ artifacts/            gitignored: bundles, cache
├─ tests/               cross-cutting tests (contracts, train/serve parity)
└─ docs/                this folder + model_card.md + threat_model.md + runbooks
```

3. Offline pipeline

3.0 Datasets: two datasets, two jobs

The PS names datasets "only as examples"; the rule is free and public.

Dataset	Job	Why this one	Feature schema
CSE-CIC-IDS2018, corrected (Liu, Engelen et al., IEEE CNS 2022)	Train, validate, test (P1, P2)	A large AWS network (420 machines, 30 servers, 50 attacker machines), 10 capture days, 7 attack scenarios with many tools (Patator, Hulk, GoldenEye, Slowloris, LOIC-HTTP/UDP, HOIC , DVWA web attacks, infiltration, Ares botnet). Labels were manually audited and the extractor bugs fixed.	CIC (fixed CICFlowMeter, ~80 features)
LUFlow (Lancaster University honeypots, labelled via threat intelligence)	Real-world showcase (P3): its own bundle, served live in the demo	Real internet attack traffic on a real university network, collected continuously since 2020, so it has real drift. Its <code>outlier</code> label ("abnormal but unexplained") is exactly what the novelty detector targets.	LUFlow (16 fields): <code>feature_spec.luflow.json</code>

Downloads: distrinet-research.be/CNS2022 (corrected 2018 + fixed CICFlowMeter) · github.com/ruzzzzz/LUFlow (or Kaggle).

2018 attack schedule (from the CIC dataset page), which drives the split design. Each attack runs for about 1 hour inside a normal working day:

Day	Attacks
Wed 14-02	FTP-BruteForce, SSH-BruteForce
Thu 15-02	DoS-GoldenEye, DoS-Slowloris
Fri 16-02	DoS-SlowHTTPTest, DoS-Hulk
Tue 20-02	DDoS-LOIC-HTTP, DDoS-LOIC-UDP
Wed 21-02	DDoS-LOIC-UDP, DDoS-HOIC
Thu 22-02, Fri 23-02	Web: Brute Force, XSS, SQL Injection
Wed 28-02, Thu 01-03	Infiltration (malicious download → internal Nmap scan)
Fri 02-03	Botnet (Ares)

Size: the original release has about 16M flows. M1 processes the CSVs **one day at a time** (never all at once), writes float32 parquet, and builds a training sample: all attack flows (capped per class) plus a fixed-seed benign sample of a few million flows. That trains comfortably on a 16 GB laptop.

Check on first download: the corrected files must contain `Src IP`, `Dst IP` and `Timestamp` (the incident correlator and the time-blocked split need them). The original 2018 CSVs lack IPs on most days. That's one more reason to use only the corrected release.

Dataset adapters (`m1/data/adapters/{cic2018,luflow}.py`) turn each source into one canonical frame: metadata columns + features (named per the matching feature spec) + `family` + `period` (day or month). Everything downstream reads only that frame.

3.1 Data → families

- Drop from features, keep as metadata: `Flow ID`, `Src IP`, `Dst IP`, `Src Port`, `Timestamp`. `Dst Port` is an open experiment: run it both ways and document the result (B12).
- Inf/NaN: count them per column, then drop or clip them, and record the counts in `data_profile.md`. Remove exact duplicates.
- **2018 label mapping** → `AttackFamily`: FTP/SSH-BruteForce → `BruteForce` · DoS GoldenEye/Slowloris/SlowHTTPTest/Hulk → `DoS` · DDoS LOIC-HTTP/LOIC-UDP/HOIC → `DDoS` · Web Brute Force/XSS/SQL Injection → `WebAttack` · Infiltration (all stages) → `Infiltration` · Ares → `Botnet`. "Attempted" labels in the corrected set → `BENIGN`, as the dataset authors recommend. Never treat them as their own class.
- **Rare rule:** any family with < 1,000 clean training flows after cleaning is merged into `Rare`. Expect `WebAttack` to be borderline (web attacks are only hundreds of flows). Decide in M1-04 from the real counts and document the outcome.
- **LUFlow mapping:** `benign` → `BENIGN`, `malicious` → `attack` (binary only; the family is `Malicious`), `outlier` → **left out of supervised training**, kept as a separate label to measure novelty capture.
- Correlation pruning ($|p| > 0.95$, keep the more interpretable feature) on the 2018 train split only → `nscore/contracts/feature_spec.json`.

3.2 Evaluation protocol

Protocol	How	Answers	Used for
P1 time-blocked + purge (2018)	For each (day, label) group, sort by timestamp: first 70% → train, next 15% → val, last 15% → test. Drop flows within 60 s of each block boundary (purge) so a single attack session can't sit on both sides. Indices are saved to <code>data/splits/</code> so every run uses the exact same split.	How well do we detect <i>known</i> families?	Main per-class metrics, threshold tuning (val), model card
P2 LOAO (2018)	For each family F in {DoS, DDoS, BruteForce, WebAttack, Infiltration, Botnet}: remove F completely from train and val, retrain RF + IF on a fixed-size sample, then measure recall on F's test flows at the calibrated benign FPR.	Can we catch a family we've never seen?	The headline chart: RF-only recall vs fusion recall for each held-out family
P2b tool holdout (2018)	Retrain without DDoS-HOIC flows only (LOIC stays). Measure HOIC recall.	Does the model learn behaviour , not tools?	One line in <code>external</code> + a Q&A answer
P3 real-world temporal (LUFlow)	Same pipeline with <code>feature_spec.lufLOW.json</code> : train RF + IF on the earliest month(s), test month by month afterwards (recall, FPR, PSI over time). Also measure the share of <code>outlier</code> flows flagged as novel by IF vs RF.	Does it hold on real traffic, and how fast does it decay?	Real-world showcase, measured drift curve
P3b recalibration (LUFlow)	Refit only the IF + thresholds on a label-free recent benign window from a later month, then re-test that month.	Can we recover from drift without new labels?	Drift → recalibrate → recover story, demo act 5

Metrics reported per class: precision, recall, F1, **FPR**, support, one-vs-rest ROC-AUC. Overall: macro-F1, binary ROC-AUC and PR-AUC, benign FPR at the operating point. Confusion matrix saved as PNG and JSON. Probability calibration checked with a reliability plot. P2b and P3/P3b go into `EvaluationReport.external`.

Why not a day-based split: in 2018, as in 2017, each attack lives on one or two specific days. Splitting by day would leave whole families out of training. Leaving a family out on purpose is what LOAO does, deliberately and measurably.

3.3 Models

Model	Train on	Role	Notes
<code>rf_binary</code>	all train flows	p(attack)	<code>class_weight="balanced_subsample"</code> by default. SMOTE compared on train only, inside the CV folds. Tune <code>n_estimators</code> , <code>max_depth</code> , <code>min_samples_leaf</code> with time-blocked CV.
<code>rf_multiclass</code>	attack flows only	which family + probabilities per family (feed expected severity)	CIC bundles only. LUFlow has no family labels, so its bundle has <code>family_head = False</code> .
<code>iforest</code>	benign train flows only	anomaly score → percentile of benign val scores	<code>contamination</code> isn't used for thresholding. We set <code>tau_anomaly</code> as a percentile (e.g. 99.5) instead.

Thresholds (M2-07): choose `tau_binary` so that benign FPR $\leq 1.0\%$ on val, and `tau_anomaly` = the 99.5th percentile of benign val anomaly scores (adds $\leq 0.5\%$ FPR). The combined operating FPR is reported. Honest note: 1% of a large network's benign flows is still a lot of flows, which is exactly why the correlator exists (§4.3). Report **incidents per hour of replay** as well as per-flow FPR.

3.4 Explainability

- `shap.TreeExplainer` for `rf_binary` (known attacks) **and** for `iforest` (novel anomalies; `TreeExplainer` supports `IsolationForest`), so every alert can be explained.
- Each alert gets the top 5 `{feature, raw value, shap_value, benign median}`. The benign median comes from `baseline_stats.json` and lets the UI say "this flow's inter-arrival time is 4,000× shorter than normal".
- Global importance is precomputed for the Model page.
- **SHAP never changes the risk score.** One number decides the order, one explanation builds trust.
- Latency budget: < 50 ms per flow on a laptop. Cap the tree depth and count if it runs slower.

3.5 Model bundle (the offline → online contract)

```
bundle_vN/
manifest.json      BundleManifest: version, dataset, split, feature_spec sha256, file sha256s, metrics
feature_spec.json  ordered features, clip/log rules (CIC or LUFlow schema)
transformer.joblib  fitted FlowTransformer
rf_binary.joblib
rf_multiclass.joblib (absent when family_head = False)
iforest.joblib
iforest_benign_val_scores.npy for score -> percentile
thresholds.json    {"tau_binary": .., "tau_anomaly": .., "operating_fpr": ..}
label_map.json      class index -> AttackFamily
drift_reference.json quantile bins + proportions, top-15 features + reference attack rate
baseline_stats.json benign median / p95 per feature
evaluation_report.json EvaluationReport schema
confusion_matrix.png, loao.png, reliability.png
```

Bundles in the registry (each one versioned and tagged):

Bundle	Trained on	Used for
<code>netsentinel-bundle:N</code>	2018, all families	Main demo and evaluation
<code>netsentinel-demo-holdout-botnet:N</code>	2018 without Botnet	Demo act 4: "this model has never seen a botnet"
<code>netsentinel-luflow:N</code>	LUFlow earliest months	Demo act 5: real traffic
<code>netsentinel-luflow-recal:N</code>	same RF, IF + thresholds refit on a recent benign window	Demo act 5: drift recovery

The manifest's `feature_spec_sha256` stops the API from scoring CIC flows with a LUFlow bundle, and vice versa.

3.6 Registry

Azure ML workspace. `MLClient.models.create_or_update` with tags `{dataset, split, contract_version, feature_schema, macro_f1, benign_fpr, held_out}`. The API resolves `MODEL_REF=azureml:netsentinel-bundle@latest`, downloads it to `artifacts/cache/`, and verifies the sha256s. If Azure is unreachable, it falls back to the cached copy. `local:` refs are supported for development.

4. Online pipeline (FastAPI)

4.1 Endpoints (/v1 , all bodies are contract models)

Method	Path	Body / query	Returns	Notes
GET	/health		{status, model_loaded, model_version}	liveness + readiness
GET	/v1/model		ModelInfo	includes feature_schema and family_head , so the UI adapts
GET	/v1/model/evaluation		EvaluationReport	feeds the Model page
POST	/v1/flows	FlowBatch (≤ 500) · header X-API-Key	ScoreBatchResponse	validates against the loaded bundle's feature spec → transform → fuse → SHAP (non-benign only) → correlate → risk → persist
GET	/v1/incidents	status, verdict, family, level, since, limit, offset, sort=risk\ last_seen	IncidentPage	
GET	/v1/incidents/{id}		IncidentDetail	
POST	/v1/incidents/{id}/actions	AnalystActionIn · header X-Analyst	AnalystActionRecord	updates status, writes audit log
GET	/v1/incidents/{id}/brief	refresh=false	Brief	lazy, cached, 8 s timeout → template fallback
GET	/v1/drift		DriftReport	
GET	/v1/metrics		LiveMetrics	throughput, p50/p95 latency, confirmed precision, MTTA
POST	/v1/admin/reload-model	{model_ref} · admin key	ModelInfo	switches the demo between bundles (2018 → holdout → LUFlow → LUFlow-recal) and starts a fresh drift window

Mock mode (NS MOCK=1): every GET serves nscore/contracts/fixtures/*.json . This is ready on day 1 so the dashboard isn't waiting on the backend.

4.2 Scoring path for one flow

```
FlowRecord → transformer.transform_records → x          (rejects flows missing spec features: 422)
p = rf_binary.predict_proba(x)[1]
a = percentile(iforest.score_samples(x), benign_val_scores)
v = fuse(p, a, tau_binary, tau_anomaly)
if v == KNOWN_ATTACK:
    if family_head: family, probs = rf_multiclass.predict / predict_proba
    else:          family, probs = Malicious, None
    conf = p
if v == NOVEL_ANOMALY: family, probs = Unknown, None; conf = novel_confidence(a, tau_anomaly)
if v != BENIGN:
    sev = policy.expected_severity(family, probs)
    top5 = shap(explainer_for(v), x) joined with baseline_stats
    incident = correlator.upsert(meta, v, family, conf, sev, top5)
    incident.risk = policy.risk_score(v, incident.max_confidence, incident.severity, incident.flow_count)
drift_monitor.observe(x, v)          # every flow, benign included
```

4.3 Incident correlator

- Key: (src_ip, dst_ip, family). Novel anomalies use (src_ip, dst_ip, "Unknown").
- Window: an open incident absorbs a matching flow if `flow.observed_at - incident.last_seen ≤ 5 min` and the incident's status isn't `resolved` or `dismissed_fp`.
- On upsert: `flow_count += 1`, `max_confidence = max(...)`, `severity` = running mean of the flows' expected severity, `last_seen`, a running mean of |SHAP| per feature for the top 5, and a recomputed risk.
- The effect: replaying a 50k-flow DDoS-HOIC attack produces about one incident per source/destination pair. Report the flows-to-incidents ratio. It's a strong slide.

4.4 Risk engine (nscore/contracts/policy.py)

This merges the team's **Risk Scoring Pipeline** doc (docs/reference/NetSentinel_Risk_Scoring_Pipeline.pdf) with the incident and novelty design. **Confidence measures certainty, severity measures consequence**: a model can be very sure about something harmless.

```
severity    = Σ_f P(f | attack) × W[f]          expected severity over the family head's probabilities
            binary-only bundle (LUFlow) → W[Malicious] · novel anomaly → W[Unknown]
confidence  = p_attack (known attack) | novel_confidence(anomaly percentile) (novel anomaly)
burst       = 1 + 0.15 × min(1, log10(flow_count) / 3)      1 flow → 1.00 · ≥1000 flows → 1.15
risk        = min(100, round(100 × (confidence × severity × burst + novelty_bonus)))  novelty_bonus = 0.10
level       = HIGH ≥ 70 · MEDIUM 40-69 · LOW < 40
```

Family	W	Why (M5 writes the full reasoning in docs/threat_model.md)
Infiltration	1.00	Deepest stage: the attacker is already inside
Botnet	0.90	A host is likely already compromised
DoS / DDoS	0.80	Service disruption, high visibility
Rare	0.80	Merged rare families: treat as serious until triaged
Unknown (novel)	0.75 + 0.10 bonus	Unknown intent, so nothing is assumed safe
BruteForce	0.70	Active attempt to gain access
Malicious (LUFlow)	0.70	Malicious per threat intelligence, type unknown
WebAttack	0.60	Could mean an exploited app, but often fails
PortScan	0.30	Reconnaissance only (not a 2018 class; kept for live capture)

What changed vs the team doc, and why

- **Expected severity** instead of the single top-family weight. If the classifier is split 50/50 between WebAttack (0.6) and Infiltration (1.0), severity = 0.8, not a coin flip between the two extremes.
- **The burst factor (the doc's Factor 3) is in the baseline**, applied per incident through the correlator, as a multiplier in [1.00, 1.15]. A single flow gets exactly the doc's formula, so the doc's worked examples still hold (62 MEDIUM, 81 HIGH, 28 LOW, all asserted in `tests/test_contracts.py`).
- **Novel anomalies get a score**, instead of being dropped at the 0.5 gate.
- **The FPR-budget threshold replaces the fixed 0.5**.

Worked examples (from the fixtures): Infiltration, 1 flow, conf 0.81 → **81 HIGH** · DDoS-HOIC, 4,210 flows, conf 0.99 → **91 HIGH** · Botnet, 37 flows, conf 0.94 → **91 HIGH** · novel anomaly, 112 flows, conf 0.71 → **69 MEDIUM** · BruteForce, 1 flow, conf 0.88 → **62 MEDIUM** · unsure WebAttack (90/10 with Infiltration), conf 0.58 → **37 LOW**.

4.5 Database (SQLite, WAL mode; swap the URL for Azure SQL or Postgres later)

```
CREATE TABLE flows (  
  flow_id TEXT PRIMARY KEY, observed_at TEXT, received_at TEXT,  
  src_ip TEXT, dst_ip TEXT, src_port INT, dst_port INT, protocol INT,  
  features_json TEXT, verdict TEXT, p_attack REAL, anomaly_pct REAL,  
  attack_family TEXT, severity REAL, incident_id TEXT REFERENCES incidents(incident_id),  
  ground_truth TEXT, model_version TEXT, latency_ms REAL  
);  
CREATE TABLE incidents (  
  incident_id TEXT PRIMARY KEY, status TEXT DEFAULT 'new', verdict TEXT, attack_family TEXT,  
  mitre_id TEXT, mitre_name TEXT, risk_score INT, risk_level TEXT, severity REAL, max_confidence REAL,  
  flow_count INT, src_ip TEXT, dst_ip TEXT, dst_port INT, first_seen TEXT, last_seen TEXT,  
  top_features_json TEXT, brief_json TEXT, model_version TEXT,  
  acknowledged_at TEXT, updated_at TEXT  
);  
CREATE TABLE analyst_actions (  
  action_id INTEGER PRIMARY KEY AUTOINCREMENT, incident_id TEXT REFERENCES incidents(incident_id),  
  analyst TEXT NOT NULL, action TEXT NOT NULL, note TEXT, at TEXT DEFAULT CURRENT_TIMESTAMP  
);  
CREATE TABLE drift_snapshots (  
  id INTEGER PRIMARY KEY AUTOINCREMENT, computed_at TEXT, model_version TEXT, window_size INT,  
  status TEXT, max_psi REAL, report_json TEXT  
);  
CREATE INDEX ix_inc_open ON incidents(status, risk_score DESC);  
CREATE INDEX ix_inc_key ON incidents(src_ip, dst_ip, attack_family, last_seen);
```

Benign flows: keep at most N recent ones (configurable), because drift only needs the rolling window.

4.6 Drift monitor

- Rolling window of the last 2,000 transformed flows. Every 500 flows it computes PSI for each of the top-15 features against the **loaded bundle's** `drift_reference.json`, plus the predicted attack rate compared with the reference rate.
- Status: max PSI < 0.10 is `ok`, < 0.25 is `watch`, anything higher is `alert`. Each check writes a snapshot. Alert status shows a "consider recalibrating" banner linked to `docs/runbook_retrain.md`.
- Within 2018 the benign traffic is scripted, so expect little drift there. **The real drift story is LUFlow**: replaying a later month through the LUFlow bundle pushes PSI to `alert`, and the recalibrated bundle brings it back.

4.7 Brief service (Azure OpenAI)

- Input: structured incident JSON only (family, verdict, confidence band, risk level, flow count, top features with raw values and benign medians, MITRE ID, IPs and port). No raw payloads.
- System prompt (extends Doc B §6): write 3–4 sentences covering what was seen, why it was flagged (only from the given features), and one suggested next step. Don't invent details. **Hedge to match the confidence band.** For `novel_anomaly`, say plainly that no known family matched. For `Malicious` (LUFlow), don't guess a type.
- Generated when first requested, then cached in `incidents.brief_json`. Times out after 8 s and falls back to a deterministic template brief (`source: "template"`). A failed brief never blocks the queue.
- Config comes only from environment variables: `AZURE_OPENAI_ENDPOINT`, `AZURE_OPENAI_API_KEY`, `AZURE_OPENAI_DEPLOYMENT`, `AZURE_OPENAI_API_VERSION`.

4.8 Security and ops of the system itself

- `X-API-Key` on ingest, `X-Analyst` (name + role) on actions, admin key on reload. Secrets live in `.env`, never in git, and CI runs secret scanning.
- Structured JSON logs with a request ID. p50/p95 latency is exposed in `/v1/metrics`.
- Pydantic validation rejects malformed flows with a 422 that names the missing feature.

5. SOC console (Streamlit)

Page	Shows	Key interactions
Live Queue	Incidents sorted by risk score . HIGH red, MEDIUM amber, LOW grey. Chips for <code>Known attack</code> and <code>Novel anomaly</code> . Header counters: open HIGHs, novel anomalies, flows/sec. Auto-refreshes every 3 s and highlights new rows. A badge shows which bundle is loaded (2018 / LUFlow).	filter by status/verdict/family/level, click through to detail
Incident Detail	Risk breakdown (confidence × severity × burst = score) · SHAP bar chart · "vs normal" table (value vs benign median) · MITRE badge · flow metadata · brief panel (loading, LLM or template label) · action timeline	Acknowledge / Escalate / Confirm / Dismiss as FP, with a note
Model & Evaluation	Model version and registry ref · per-class table · confusion matrix · LOAO chart · Real-world panel (LUFlow month-by-month + recalibration, from <code>external</code>) · operating FPR · limitations	—
Drift & Health	PSI per feature against the 0.10/0.25 lines · status banner · throughput · latency	—
Analyst Metrics	Analyst-confirmed precision · FP dismiss rate · MTTA · actions per analyst	—

When `family_head` is false (LUFlow bundle), the UI hides family-specific widgets (MITRE, confusion matrix) instead of showing empty ones. Analyst sign-in is a name plus a role in session state, sent as `X-Analyst`. React stays a stretch goal and uses the same API.

6. Demo storyline (≈ 5 min)

1. **Calm.** 2018 benign replay runs, the queue is empty, drift is `ok`, and the model page shows v N from Azure ML.
2. **Known attack.** A brute-force burst arrives as one incident, mapped to MITRE T1110. Open it and show the **risk breakdown** ("88% sure × severity 0.7 × burst"). Generate the brief, then Escalate.

3. **Alert storm.** Replay DDoS-HOIC: thousands of flows become **one HIGH incident**. "That's the alert-fatigue answer."
4. **The novel attack.** Reload the **holdout bundle** (it has never seen a botnet) and replay Ares botnet traffic. The RF stays quiet and the Isolation Forest fires, so it appears as **Novel anomaly** with a SHAP explanation. Show the LOAO chart: "measured across every family, not just this demo".
5. **Real traffic.** Reload the **LUFlow bundle** and replay real honeypot traffic from a **later month**. Real attackers show up as incidents, unexplained `outlier` traffic gets flagged as novel, and **drift jumps to alert** because the internet changed since training. Reload **LUFlow-recal** (refit on a recent benign window, no labels needed) and drift drops back to `ok`. Show the month-by-month chart.
6. **Honesty.** Dismiss a low-confidence incident as a false positive and the analyst-confirmed precision updates live. Walk through the limitations slide.
7. Fallbacks: a recorded video of acts 1–6. Briefs fall back to the template if Azure OpenAI is unreachable. Bundles load from the local cache if Azure ML is unreachable.

7. Tech stack

Layer	Choice	Why
ML	scikit-learn, imbalanced-learn, shap, pandas/pyarrow	It's what the PS names, it's fast on a laptop, and TreeSHAP is exact and quick
Registry	Azure ML (SDK v2) + local cache	Real use of Microsoft tooling, with a demo fallback
API	FastAPI, Uvicorn, Pydantic v2, SQLAlchemy Core / sqlite3	Contract-typed and quick to build
LLM	Azure OpenAI (gpt-4o-mini class deployment)	Cheap, quick, and a Microsoft service
UI	Streamlit + Plotly	Fast to build. React is a stretch goal
Infra	Docker Compose (api + dashboard). Optional deploy to Azure Container Apps	One command runs the demo
CI	GitHub Actions: ruff, pytest, gitleaks	Required checks on PRs

8. Requirements traceability

PS requirement	Where it's met	Evidence on demo day
Normal vs attack + attack types	<code>rf_binary</code> , <code>rf_multiclass</code>	Family label on every 2018 incident
Surfaces novel attacks	<code>iforest</code> + <code>fusion.fuse</code>	Demo act 4 + LOAO chart + LUFlow outlier capture (act 5)
Precision / recall / FPR / AUC	<code>ml/evaluate</code> → <code>EvaluationReport</code>	Model page, model card
Class imbalance	class weights, SMOTE comparison, Rare rule	Model card section + before/after table
Discuss model drift	<code>nscore/drift</code> , <code>/v1/drift</code> , drift page, P3/P3b, model card §Drift	Measured decay on real traffic (LUFlow month by month), live drift alert and recovery in act 5
Alert SOC, no auto-block	incidents + risk engine + actions + audit log	Demo acts 2–6
Honest evaluation	purged time-blocked split, FPR-budget thresholds, corrected dataset, limitations	Model card, Q&A sheet

PS requirement	Where it's met	Evidence on demo day
Works beyond lab data	LUFlow bundle	Act 5

9. Key decisions (short ADRs)

#	Decision	Alternatives considered	Why
ADR-1	RF + benign-only Isolation Forest fusion	RF only; autoencoder	RF only can't address "novel". An autoencoder is slower to train and to explain. IF has TreeSHAP support and trains in seconds.
ADR-2	Purged time-blocked per-class split + LOAO	day-based split; random split	A day-based split removes whole families from training. A random split leaks sessions.
ADR-3	Incidents, not per-flow alerts	per-flow alerts	Per-flow alerts make alert fatigue worse, which contradicts the pitch.
ADR-4	Contract-first monorepo with shared <code>nscore</code>	separate repos; no contracts	Lets 5 people work in parallel and avoids train/serve skew.
ADR-5	Streamlit first	React first	Hackathon speed. Both call the same API.
ADR-6	SQLite (WAL)	Postgres	Zero setup. Moving later is just a URL change.
ADR-7	Rebuild SHAP explainers at load	pickle explainers	Pickled explainers break across shap versions. Rebuilding costs milliseconds.
ADR-8	Corrected CSE-CIC-IDS2018 for train/val/test; LUFlow as a live real-traffic showcase with its own bundle	CIC-IDS2017; 2017 + 2018 cross-network; NetFlow-v3 family; UNSW-NB15	2018 is the larger, more varied labelled network with more attack tools and rare-class examples. The labels are audited. LUFlow adds real traffic and real drift , which 2018's scripted benign traffic lacks. One labelled dataset keeps M1's workload realistic. Trade-off: no same-schema cross-network test (accepted).
ADR-9	Risk = confidence × expected severity × burst (+ novelty bonus), 3 levels	raw confidence; per-flow scores; P1–P4 bands	Adopts the team's risk doc (confidence ≠ consequence, 3 levels, defensible weights) and adds expected severity, incident-level burst and novelty scoring.

Source: docs/03_architecture.md · github.com/Saroy-ctrl/netsentinel